Lab 6: Code and Branch Coverage Analysis

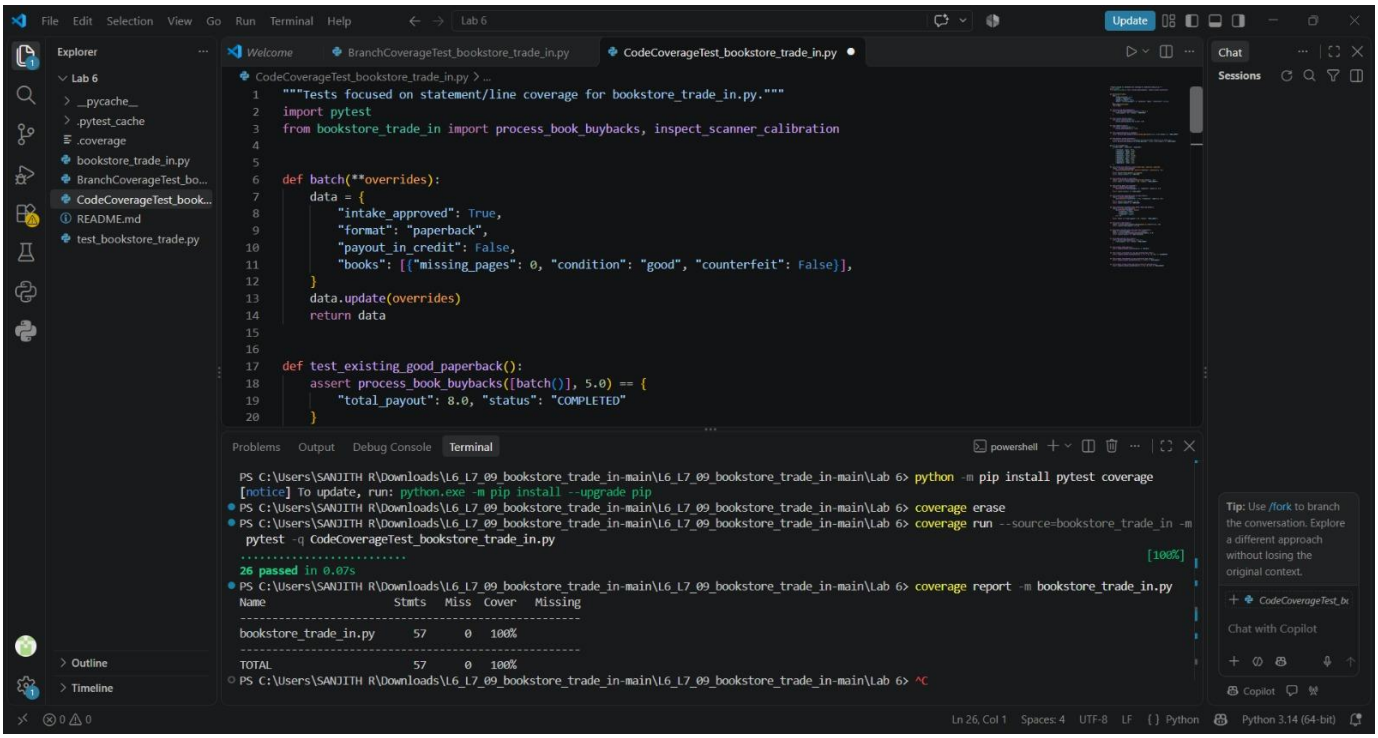
Bookstore Trade-In Application

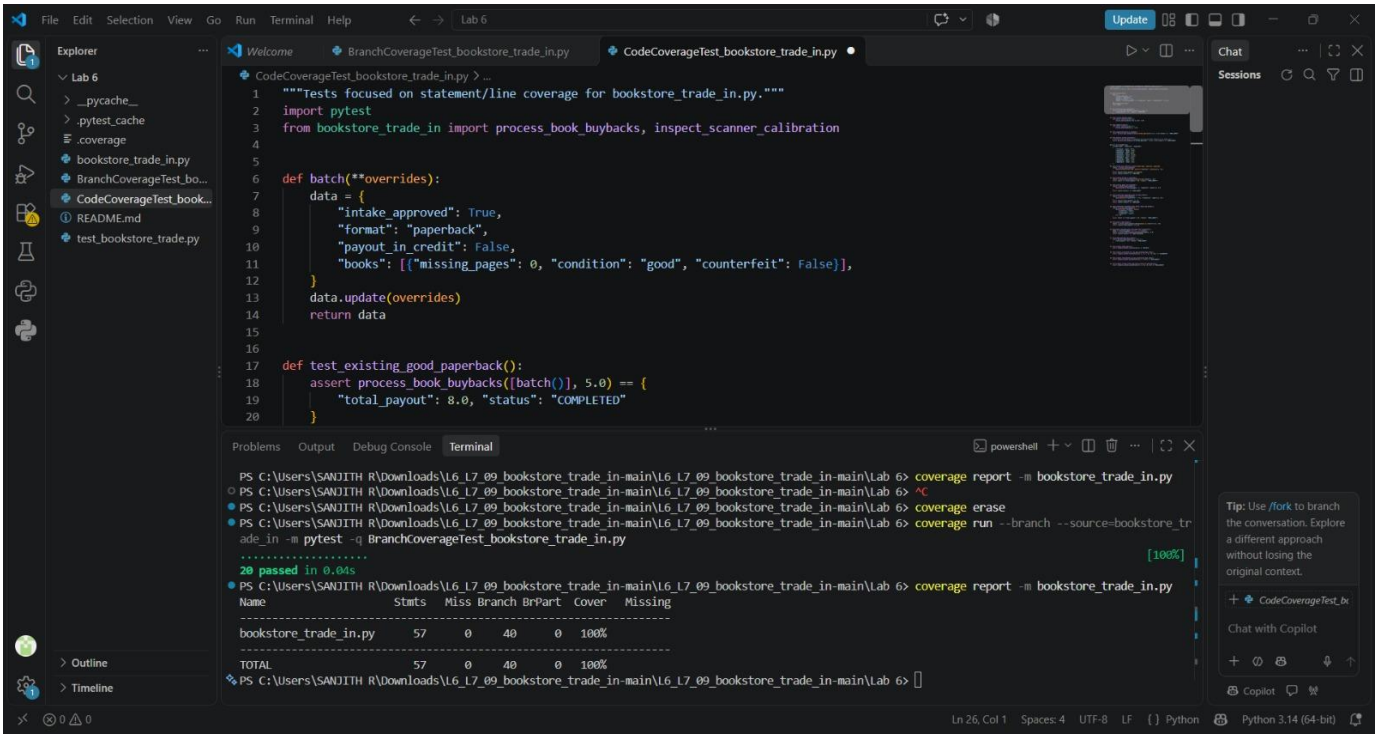
Repository: SETAPESU26/L6_L7_09_bookstore_trade_in

1. Objective

Measure and analyse line and branch coverage for the Bookstore Trade-In application, add tests for different conditions and edge cases, and review the resulting coverage reports.

2. Screen shots





Summary: The captured terminal outputs show all tests passing. Code coverage reports 57 statements, 0 missing, and 100% coverage. Branch coverage reports 40 branches, 0 partial branches, and 100% coverage.

4. Commands Used

```
python -m pip install pytest coverage
coverage erase
coverage run --source=bookstore_trade_in -m pytest -q CodeCoverageTest_bookstore_trade_in.py
coverage report -m bookstore_trade_in.py
coverage erase
coverage run --branch --source=bookstore_trade_in -m pytest -q BranchCoverageTest_bookstore_trade_in.py
coverage report -m bookstore_trade_in.py
```

5. Terminal Screenshots

The two terminal screenshots are submitted as separate files: Code_coverage.jpeg (26 tests passed; 100% line coverage) and Branch_coverage.png (20 tests passed; 100% coverage; 40 branches and 0 partial branches).

6. GitHub Commits

Code coverage: Add code coverage tests and report
https://github.com/sanjithramanjaneya-source/L6_L7_09_bookstore_trade_in/commit/75d535d0e2f695d82450a8fbd625aa54c69f5fc8

Branch coverage: Add branch coverage tests and report
https://github.com/sanjithramanjaneya-source/L6_L7_09_bookstore_trade_in/commit/3fa967924e27487e9c86522e75869fce8bd1100e

7. Conclusion

Based on the captured terminal outputs, the added tests achieved 100% line coverage and 100% branch coverage, exceeding the lab requirement of at least 90%. The two test files, two terminal screenshots, and two separate commits form the primary deliverables.

GIT HUB LINK:

https://github.com/sanjithramanjaneya-source/L6_L7_09_bookstore_trade_in.git